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***Title**	Title / Name (20 Words Max) * Echo Verse: Where the Audio Meets Intelligence **Total Number of words: 0 / 20**
***Developed as part of**	Academic Requirement/Study Project
***Choose the Financial Year, during the Idea-PoC/Innovation Developed**	2024-25
***Sector / Domain**	Theme * ICT, cyber-physical systems, Blockchain, Cognitive computing, Cloud co…
***Innovation Type**	Service
***Development Stage - Technology Maturity of the Solution/Innovation in terms of Technology Readiness Level TRL (if applicable** (Refer TRL Stages)**)**	TRL 3 : Applied research. First laboratory tests completed; proof of concept
***Define the problem and its relevance to today's market / sociaty / industry need (Max: 100 Words)**	

	Define the problem and its relevance to today's market / society / industry need * addresses this gap by integrating visual (spectrogram) and textual (transcription) features using deep learning for more accurate sound understanding. This is highly relevant in today's market, where intelligent audio interpretation is critical for safety (disaster alerts), accessibility (speech-to-text), and automation (smart homes, voice assistants). EchoVerse offers a **Total Number of words: 0 / 100**
***Describe the Solution / Proposed / Developed (Max: 100 Words)**	Describe the Solution / Proposed / Developed * EchoVerse is a deep learning-based system that intelligently classifies audio by combining two types of features: visual (from spectrogram images) and textual (from speech transcriptions). The audio is first converted into a spectrogram and processed using a ResNet50 model, while the transcribed text is analyzed using a BiLSTM network with an attention mechanism. These features are then **Total Number of words: 0 / 100**
***Explain the uniqueness and distinctive features of the (product / process / service) solution (Max: 100 Words)**	Explain the uniqueness and distinctive features of the (product / process / service) solution * EchoVerse stands out by using a multimodal approach—combining both spectrogram images and speech text to classify audio more accurately than traditional models. Unlike typical systems that rely only on sound patterns, EchoVerse understands both the acoustic signals and the contextual meaning of audio. It integrates powerful deep learning models like ResNet50 and BiLSTM **Total Number of words: 0 / 100**
***How your proposed / developed (product / process / service) solution is different from similiar kind of product by the competitors if any (Max: 100 Words)**	How your proposed / developed (product / process / service) solution is different from similiar kind of … EchoVerse is different from other audio systems because it uses both sound images (spectrograms) and text (from speech) together, while most other tools use only one. This makes it more accurate, especially in noisy or unclear situations. It uses advanced deep learning models (ResNet50 and BiLSTM with attention) to understand not just what was heard, but also what was said. **Total Number of words: 0 / 100**
***Is there any IP or Patentable Component associated with the Solution?**	No
***Has the Solution Received any Innovation Grant/Seefund Support?**	No
***Are there any Recognitions (State/National/International) Obtained by the Solution?**	No
***Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?**	No
***Had the Solution Received any Pre-Incubation/Incubation Support?**	No
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